Memo: Whale Position and Velocity Estimation

MANTA Simulator

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1 Purpose

The estimator receives occasional whale-position measurements formed by triangulating two vehicle bearings. It combines each measured position with a motion model to estimate the whale's north/east position and velocity. Odyssey projects the latest estimate forward between measurements so a vehicle can steer toward the predicted whale location.

The ROS whale-estimator node runs a Kalman filter when a new triangulated position arrives. Odyssey then predicts where the whale would be at the current control time using the latest estimate and the same constant-rate return model.

The current configuration sets the north baseline velocity to 1.5 m/s, the east baseline to 0 m/s, and the velocity return time to 1800 s. Therefore the model does not bring the whole velocity vector to zero: north tends toward 1.5 m/s, while east tends toward zero. This is an estimator assumption; it does not change the whale simulator's actual trajectory.

2 State for one axis

North and east are filtered independently using the same equations. For one axis, the state is

$$\mathbf{x} = \begin{bmatrix} p \\ v \end{bmatrix},$$

where p is position in metres and v is velocity in metres per second. The filter also stores

$$\mathbf{P} = \begin{bmatrix} P_{pp} & P_{pv} \\ P_{pv} & P_{vv} \end{bmatrix}.$$

The diagonal entries are position and velocity variances. The off-diagonal entry is their covariance, describing how position and velocity errors vary together. A triangulated position supplies measured position z and measurement variance R; it does not directly measure velocity.

3 Motion model: constant-rate return toward a baseline

Let v_0 be the configured baseline velocity and τ the return duration. At the start of a prediction interval, let v be the estimated velocity before seeing the current prediction, and $v_{prev} - v_0$ its difference from baseline. The model removes that starting difference at a constant rate, reaching baseline after τ seconds if no newer position estimate arrives:

$$v_{pred} = v_0 + (v_{prev} - v_0)\max(1 - \Delta t/\tau, 0). \tag{1}$$

For $0 \le t < \tau$, velocity changes along a straight line. At and after τ, it remains at v_0. Position is the integral of that changing velocity:

$$p_{pred} = \begin{cases} p(0) + v_0\Delta t + (v_{prev} - v_0)\left(\Delta t - \dfrac{\Delta t^2}{2\tau}\right), & 0 \le \Delta t < \tau, \\ p(0) + v_0 t + (v_{prev} - v_0)\dfrac{\tau}{2}, & t \ge \tau. \end{cases} \tag{2}$$

This keeps position continuous and accounts for travel while velocity returns to baseline. Unlike an exponential return, the linear rule reaches baseline in finite time.

4 Discrete Kalman prediction

For a measurement interval Δt, define the remaining fraction of the starting velocity difference and the integrated travel factor:

$$a = \max(1 - \Delta t/\tau, 0), \qquad T = \begin{cases} \Delta t - \dfrac{\Delta t^2}{2\tau}, & \Delta t < \tau, \\ \tau/2, & \Delta t \ge \tau. \end{cases}$$

The state prediction using the model is

$$\mathbf{x}^- = \mathbf{F}\mathbf{x} + \mathbf{B}\mathbf{u}, \qquad \mathbf{F} = \begin{bmatrix} 1 & T \\ 0 & a \end{bmatrix}, \qquad \mathbf{B}\mathbf{u} = \begin{bmatrix} v_0(\Delta t - T) \\ v_0(1-a) \end{bmatrix}.$$

$\mathbf{F}$ propagates position and velocity deviations; the control term adds back baseline motion. Equivalently,

$$p^- = p + v_0\Delta t + (v - v_0)T, \qquad v^- = v_0 + (v - v_0)a.$$

If the interval lasts at least τ, the starting velocity difference has fully returned to baseline.

5 Process noise: origin of Q

The process-noise covariance $\mathbf{Q}$ describes uncertainty added during one prediction interval. The model assumes random acceleration with intensity q. Over a small time step dt, acceleration causes a velocity change proportional to dt. That velocity change also affects position: a change introduced during the interval contributes to position for the remaining time. Integrating these effects over an interval of length Δt gives

$$Q_{pp} = q\frac{\Delta t^3}{3}, \qquad Q_{pv} = q\frac{\Delta t^2}{2}, \qquad Q_{vv} = q\Delta t.$$

The subscripts indicate which uncertainty is described: Q_{pp} is position variance, Q_{vv} is velocity variance, and Q_{pv} is their covariance. Together they form

$$\mathbf{Q} = \begin{bmatrix} Q_{pp} & Q_{pv} \\ Q_{pv} & Q_{vv} \end{bmatrix}, \qquad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}),$$

where $\mathbf{w}$ is the random position and velocity change modeled during the prediction interval.

The deterministic prediction handles the constant-rate return toward baseline; q describes random acceleration around that prediction. For simplicity, this Q calculation treats the random acceleration as acting on a constant-velocity state during the interval, so it is an approximation alongside the deterministic return model.

6 Measurement update

When a new triangulated whale position arrives, the estimator first calls FilterPy's `predict()` function. This advances the previous position and velocity estimates to the new measurement time using the configured baseline-return model. It also updates the filter's uncertainty.

The estimator then calls FilterPy's `update()` function with the measured position and its variance. This corrects the predicted position and velocity using the difference between the predicted and measured positions. The measurement variance controls how strongly the filter trusts the new position compared with its prediction.

For the first position, there is no previous estimate to predict from. The estimator initializes position from the measurement, velocity from the configured baseline, and their uncertainties from the measurement variance and `initial_velocity_variance_m2ps2`.

7 Velocity fit from recent positions

The estimator also fits a straight line through recent triangulated positions to estimate velocity. It waits for at least three positions, uses at most the latest five, and removes positions older than `velocity_window_max_age_s`. The slope against simulated time is

$$\hat{v} = \frac{\sum_i (t_i - \bar{t})(p_i - \bar{p})}{\sum_i (t_i - \bar{t})^2}.$$

North and east slopes are computed separately. Fit residuals contribute to the velocity variance, and a term involving q allows for acceleration during the fitting interval.

The Kalman filter has already used these same positions. Therefore the fitted velocity replaces the current velocity estimate and variance; it is not treated as a new independent Kalman measurement. The code also sets the position–velocity covariance to zero after applying the fit.

8 Odyssey prediction between measurements

Odyssey projects the latest estimate to the current simulated time using the same constant-rate return:

$$\hat{p}(t) = \begin{cases} \hat{p}(0) + v_0 t + (\hat{v}(0) - v_0)\left(t - \dfrac{t^2}{2\tau}\right), & t < \tau, \\ \hat{p}(0) + v_0 t + (\hat{v}(0) - v_0)\dfrac{\tau}{2}, & t \geq \tau, \end{cases}$$

$$\hat{v}(t) = v_0 + (\hat{v}(0) - v_0)\max(1 - t/\tau, 0).$$

The trailing behavior steers toward the projected position. Under the current configuration, projected north velocity approaches 1.5 m/s and east velocity approaches zero. If both baseline components were zero, the estimated velocity vector would gradually approach zero instead.

9 Configuration values

Parameter	Meaning
`q`	Acceleration-noise intensity. Larger values add more uncertainty between measurements. Units: m^2/s^3.
`initial_velocity_variance_m2ps2`	Uncertainty in starting velocity; separate from q.
`whale_velocity_return_time_s`	Time for the starting velocity difference to return linearly to baseline.
`whale_baseline_velocity_north_mps`, `whale_baseline_velocity_east_mps`	Long-term north/east velocity assumptions.
`velocity_window_max_age_s`	Maximum age of positions in the recent-position velocity fit.
R	Variance of an individual triangulated position; controls how strongly it affects the Kalman update.

10 Assumptions and limitations

The model assumes smooth velocity changes around a configured baseline with Gaussian acceleration noise. The baseline is a prediction assumption, not a command to the simulated whale. North and east are filtered independently, so their motion correlation is not represented. The recent-position velocity fit uses the same observations as the Kalman filter and overwrites the velocity estimate rather than adding an independent observation.

References

[1] D. S. Johnson, J. M. London, M.-A. Lea, and J. W. Durban, "Continuous-time correlated random walk model for animal telemetry data," *Ecology*, vol. 89, no. 5, pp. 1208–1215, 2008. doi:10.1890/07-1032.1.